

SDC Assignment 2 - ITRI Dataset Visualization

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Spring 2026

Task 1: Rosbag Information

```
root@laptop:/ros_ws/data# rosbag info /ros_ws/data/guanfu_road_clipped.bag
path:      /ros_ws/data/guanfu_road_clipped.bag
version:   2.0
duration:  60.0s
start:     Sep 25 2024 03:14:50.00 (1727234090.00)
end:       Sep 25 2024 03:15:49.00 (1727234150.00)
size:      7.6 GB
messages:  249147
compression: none [7501/7501 chunks]
types:
  can_msgs/Frame [64ae5cebf967dc6aae4e78f5683a5b25]
  dbw_pacifica_msgs/AcceleratorPedalCmd [a1a82e02e150ecb7c85bb191a9e67d46]
  dbw_pacifica_msgs/AcceleratorPedalReport [79e5f58f94a8fa77ed7d602d29d0230c]
  dbw_pacifica_msgs/ActuatorsReport [d66d15b164e1cf2d467650778182b722]
  dbw_pacifica_msgs/BrakeCmd [df5046f02deb0972bd5ea27261595bc6]
  dbw_pacifica_msgs/BrakeReport [c23e437775db686330778b5ba5b0b41b]
  dbw_pacifica_msgs/DriverInputReport [7d92e8189302532750bc994bbf345e99]
  dbw_pacifica_msgs/GearCmd [0e1ada801a24754a8b6ec38e04e82ab5]
  dbw_pacifica_msgs/GearReport [82edf69509597cc6f96e7ba84e5d0077]
  dbw_pacifica_msgs/GlobalEnableCmd [6bed8b1974abe40f81a2b6c9d5a493f6]
  dbw_pacifica_msgs/LowVoltageSystemReport [c352ea40227e6ba399244c0eeed0064c]
  dbw_pacifica_msgs/MiscCmd [c5a8f076a15a9659536c514cdcd82275]
  dbw_pacifica_msgs/MiscReport [07d2b659403a0cbd241b5457fee2e270]
  dbw_pacifica_msgs/Steering2Report [009145df632b20263fd61a2cbd47cc87]
  dbw_pacifica_msgs/SteeringCmd [01afefccdc79bc13944f883510af8807]
  dbw_pacifica_msgs/SteeringReport [3d804816a4b20624da24c6e1d360c70b]
  dbw_pacifica_msgs/SurroundReport [7fb6baa192c10e400b9071053bd9c683]
  dbw_pacifica_msgs/TirePressureReport [a2c91f746e5d8bec139c834f92ac7468]
  dbw_pacifica_msgs/WheelPositionReport [dcd40156270607476a632a433cc6d45]
  dbw_pacifica_msgs/WheelSpeedReport [7f27720c7c80e612329b2205902e5bba]
  diagnostic_msgs/DiagnosticArray [60810da900de1dd6ddd437c3503511da]
  geometry_msgs/PoseStamped [d3812c3cbc69362b77dc0b19b345f8f5]
  geometry_msgs/Vector3Stamped [7b324c7325e683bf02a9b14b01090ec7]
  itri_msgs/CarState [e29dfa68c85fb4e233bbfc19ecfc0749]
  itri_msgs/DetectedObjectArray [276d607237a4270f49c5de8f71694bfa]
  itri_msgs/NdtStatistics [b21481a49aaae81584b34de5ce8cef65]
  itri_msgs/PredictState [72d0803d1465f548e6ee4eb9d0ba77d3]
  itri_msgs/ProgramOutput [a0307b07c162469a5f8993c1679034b8]
  itri_msgs/VehicleState [356941fd514ebcd703629e85894a6315]
  jsk_recognition_msgs/PolygonArray [709b37d39871cfdbbfbd5c41cf9bc2be]
  nmea_msgs/Sentence [9f221efc5f4b3bac7ce4af102b32308b]
  sensor_msgs/CompressedImage [8f7a12909da2c9d3332d540a0977563f]
  sensor_msgs/Image [060021388200f6f0f447d0fcd9c64743]
  sensor_msgs/Imu [6a62c6daae103f4ff57a132d6f95cec2]
```

Figure 1: Bag Info Part 1

topics:	/Navtech/Polar	240 msgs	: sensor_msgs/Image
	/accelerator_pedal_cmd	2400 msgs	: dbw_pacifica_msgs/AcceleratorPedalCmd
	/accelerator_pedal_report	6000 msgs	: dbw_pacifica_msgs/AcceleratorPedalReport
	/actuator_report	600 msgs	: dbw_pacifica_msgs/ActuatorsReport
	/actuators_cmd	2400 msgs	: dbw_pacifica_msgs/MiscCmd
	/ars548/radar_front/detections	1196 msgs	: sensor_msgs/PointCloud2
	/ars548/radar_front/objects	1196 msgs	: sensor_msgs/PointCloud2
	/ars548/radar_rear/detections	1198 msgs	: sensor_msgs/PointCloud2
	/ars548/radar_rear/objects	1197 msgs	: sensor_msgs/PointCloud2
	/black_panther_iostat/output	54 msgs	: itri_msgs/ProgramOutput
	/black_panther_iotop/output	60 msgs	: itri_msgs/ProgramOutput
	/black_panther_meminfo/output	60 msgs	: itri_msgs/ProgramOutput
	/black_panther_nvidia_smi/output	60 msgs	: itri_msgs/ProgramOutput
	/black_panther_ps/output	30 msgs	: itri_msgs/ProgramOutput
	/black_panther_top/output	30 msgs	: itri_msgs/ProgramOutput
	/brake_cmd	2400 msgs	: dbw_pacifica_msgs/BrakeCmd
	/brake_report	6000 msgs	: dbw_pacifica_msgs/BrakeReport
	/car_state	6000 msgs	: itri_msgs/CarState
	/compensate_point	600 msgs	: sensor_msgs/PointCloud2
	/driver_input_report	600 msgs	: dbw_pacifica_msgs/DriverInputReport
	/filter/velocity	600 msgs	: geometry_msgs/Vector3Stamped
	/filtered_cluster_hulls	600 msgs	: jsk_recognition_msgs/PolygonArray
	/filtered_cluster_hulls_F	600 msgs	: jsk_recognition_msgs/PolygonArray
	/filtered_cluster_hulls_L	600 msgs	: jsk_recognition_msgs/PolygonArray
	/filtered_cluster_hulls_R	600 msgs	: jsk_recognition_msgs/PolygonArray
	/fix_vs330	600 msgs	: sensor_msgs/NavSatFix
	/front_cluster_hulls	600 msgs	: jsk_recognition_msgs/PolygonArray
	/front_detection_objects	600 msgs	: itri_msgs/DetectedObjectArray
	/front_detection_points	600 msgs	: sensor_msgs/PointCloud2
	/front_lidar_objects	600 msgs	: itri_msgs/DetectedObjectArray
	/front_lidar_objects_raw	600 msgs	: itri_msgs/DetectedObjectArray
	/front_lidar_points	600 msgs	: sensor_msgs/PointCloud2
	/gear_cmd	2400 msgs	: dbw_pacifica_msgs/GearCmd
	/gear_report	600 msgs	: dbw_pacifica_msgs/GearReport
	/global_enable_cmd	2400 msgs	: dbw_pacifica_msgs/GlobalEnableCmd
	/gnss	240 msgs	: sensor_msgs/NavSatFix
	/imu/data	6000 msgs	: sensor_msgs/Imu
	/imu/time_ref	6480 msgs	: sensor_msgs/TimeReference
	/innovusion/fk/txyzi	600 msgs	: sensor_msgs/PointCloud2
	/left_cluster_hulls	600 msgs	: jsk_recognition_msgs/PolygonArray
	/left_detection_objects	600 msgs	: itri_msgs/DetectedObjectArray
	/left_detection_points	600 msgs	: sensor_msgs/PointCloud2
	/left_lidar_objects	600 msgs	: itri_msgs/DetectedObjectArray
	/left_lidar_objects_raw	600 msgs	: itri_msgs/DetectedObjectArray
	/left_lidar_points	600 msgs	: sensor_msgs/PointCloud2
	/local_pose	6000 msgs	: geometry_msgs/PoseStamped
	/localizer_node/gnss_pose	240 msgs	: geometry_msgs/PoseStamped
	/localizer_node/log	4440 msgs	: std_msgs/String

Figure 2: Bag Info Part 2

/lucid_cameras_x01/gige_100_fr_hdr/compressed	1667	msgs	: sensor_msgs/CompressedImage
/lucid_cameras_x01/gige_60_bl/compressed	2119	msgs	: sensor_msgs/CompressedImage
/lucid_cameras_x01/gige_60_br/compressed	2120	msgs	: sensor_msgs/CompressedImage
/misc_report	6000	msgs	: dbw_pacifica_msgs/MiscReport
/monitor/cpu_usage	28	msgs	: std_msgs/UInt16MultiArray
/monitor/gpu_usage	57	msgs	: std_msgs/UInt16MultiArray
/monitor/major_system	600	msgs	: diagnostic_msgs/DiagnosticArray
/monitor/major_topics	58	msgs	: diagnostic_msgs/DiagnosticArray
/monitor/minor_system	600	msgs	: diagnostic_msgs/DiagnosticArray
/monitor/minor_topics	57	msgs	: diagnostic_msgs/DiagnosticArray
/nmea_sentence	1800	msgs	: nmea_msgs/Sentence
/no_ground_front	600	msgs	: sensor_msgs/PointCloud2
/no_ground_left	600	msgs	: sensor_msgs/PointCloud2
/no_ground_right	600	msgs	: sensor_msgs/PointCloud2
/no_ground_top	600	msgs	: sensor_msgs/PointCloud2
/ouster/back_991939001246	600	msgs	: sensor_msgs/PointCloud2
/ouster/back_991939001246/imu	6000	msgs	: sensor_msgs/Imu
/ouster/front_991923000849	600	msgs	: sensor_msgs/PointCloud2
/ouster/front_991923000849/imu	5999	msgs	: sensor_msgs/Imu
/ouster/top_122219002200	600	msgs	: sensor_msgs/PointCloud2
/ouster/top_122219002200/imu	6000	msgs	: sensor_msgs/Imu
/received_dbw_messages	48994	msgs	: can_msgs/Frame
/right_cluster_hulls	600	msgs	: jsk_recognition_msgs/PolygonArray
/right_detection_objects	600	msgs	: itri_msgs/DetectedObjectArray
/right_detection_points	600	msgs	: sensor_msgs/PointCloud2
/right_lidar_objects	600	msgs	: itri_msgs/DetectedObjectArray
/right_lidar_objects_raw	600	msgs	: itri_msgs/DetectedObjectArray
/right_lidar_points	600	msgs	: sensor_msgs/PointCloud2
/sent_dbw_messages	14400	msgs	: can_msgs/Frame
/steering_2_report	6000	msgs	: dbw_pacifica_msgs/Steering2Report
/steering_cmd	2400	msgs	: dbw_pacifica_msgs/SteeringCmd
/steering_report	6000	msgs	: dbw_pacifica_msgs/SteeringReport
/surround_report	600	msgs	: dbw_pacifica_msgs/SurroundReport
/tf	20669	msgs	: tf2_msgs/TFMessage
/tf_static	1	msg	: tf2_msgs/TFMessage
/tire_pressure_report	120	msgs	: dbw_pacifica_msgs/TirePressureReport
/top_cluster_hulls	600	msgs	: jsk_recognition_msgs/PolygonArray
/top_detection_objects	600	msgs	: itri_msgs/DetectedObjectArray
/top_detection_points	600	msgs	: sensor_msgs/PointCloud2
/top_lidar_objects	600	msgs	: itri_msgs/DetectedObjectArray
/top_lidar_objects_raw	600	msgs	: itri_msgs/DetectedObjectArray
/top_lidar_points	600	msgs	: sensor_msgs/PointCloud2
/vehicle_state	6000	msgs	: itri_msgs/VehicleState
/velodyne/left_vlp16	600	msgs	: sensor_msgs/PointCloud2
/velodyne/right_vlp16	600	msgs	: sensor_msgs/PointCloud2
/wheel_position_report	600	msgs	: dbw_pacifica_msgs/WheelPositionReport
/wheel_speed_report	3000	msgs	: dbw_pacifica_msgs/WheelSpeedReport

Figure 3: Bag Info Part 3

Task2: Point Cloud Topic Information

```

root@laptop:/ros_ws/src/pc_profiler_node
X range: [-82.4, 156.9] m, Y range: [-19.0, 49.0] m, Z range: [-2.7, 18.4] m

Topic Name: /ouster/top_122219002200
Timestamp: 1727234104.805
Total number of points: 21134
X range: [-80.2, 156.2] m, Y range: [-18.8, 60.5] m, Z range: [-2.7, 18.4] m

Topic Name: /ouster/top_122219002200
Timestamp: 1727234104.905
Total number of points: 21122
X range: [-83.8, 155.4] m, Y range: [-20.3, 56.8] m, Z range: [-2.6, 18.7] m

Topic Name: /ouster/top_122219002200
Timestamp: 1727234105.005
Total number of points: 20955
X range: [-84.7, 154.7] m, Y range: [-17.6, 60.5] m, Z range: [-2.5, 15.8] m

Topic Name: /ouster/top_122219002200
Timestamp: 1727234105.105
Total number of points: 20868
X range: [-85.5, 153.9] m, Y range: [-18.0, 91.7] m, Z range: [-2.9, 18.3] m

[RUNNING] Bag Time: 1727234105.179410 Duration: 15.179176 / 59.999319
[RUNNING] Bag Time: 1727234105.179507 Duration: 15.179273 / 59.999319
[RUNNING] Bag Time: 1727234105.179515 Duration: 15.179281 / 59.999319
[RUNNING] Bag Time: 1727234105.180116 Duration: 15.179882 / 59.999319
[RUNNING] Bag Time: 1727234105.180199 Duration: 15.179965 / 59.999319
[RUNNING] Bag Time: 1727234105.180260 Duration: 15.180026 / 59.999319
[RUNNING] Bag Time: 1727234105.180953 Duration: 15.180718 / 59.999319
[RUNNING] Bag Time: 1727234105.181086 Duration: 15.180852 / 59.999319
[RUNNING] Bag Time: 1727234105.181290 Duration: 15.181055 / 59.999319
[RUNNING] Bag Time: 1727234105.181587 Duration: 15.181352 / 59.999319
[RUNNING] Bag Time: 1727234105.181790 Duration: 15.181556 / 59.999319
[RUNNING] Bag Time: 1727234105.181895 Duration: 15.181661 / 59.999319
[RUNNING] Bag Time: 1727234105.181973 Duration: 15.181739 / 59.999319
[RUNNING] Bag Time: 1727234105.182587 Duration: 15.182352 / 59.999319
[RUNNING] Bag Time: 1727234105.184960 Duration: 15.184725 / 59.999319
[RUNNING] Bag Time: 1727234105.186220 Duration: 15.185986 / 59.999319
[RUNNING] Bag Time: 1727234105.186406 Duration: 15.186171 / 59.999319
[RUNNING] Bag Time: 1727234105.187211 Duration: 15.186976 / 59.999319
[RUNNING] Bag Time: 1727234105.187301 Duration: 15.187067 / 59.999319
[RUNNING] Bag Time: 1727234105.188253 Duration: 15.188018 / 59.999319
[RUNNING] Bag Time: 1727234105.188329 Duration: 15.188095 / 59.999319
[RUNNING] Bag Time: 1727234105.189096 Duration: 15.188862 / 59.999319

Done checking log file disk usage. Usage is <1GB.

started roslaunch server http://laptop:46519/
ros_comm version 1.17.4

SUMMARY
=====

PARAMETERS
* /roscdistro: noetic
* /rosversion: 1.17.4

NODES

auto-starting new master
process[master]: started with pid [477]
ROS_MASTER_URI=http://laptop:11311/

setting /run_id to a6b02ba0-2344-11f1-a68f-4845e6127e59
process[rosout-1]: started with pid [487]
started core service [/rosout]

```

Figure 4: Terminal output of pc_profiler.py for LiDAR (/ouster/top_122219002200)

```

root@laptop:/ros_ws/src/pc_profiler_node
X range: [3.3, 298.1] m, Y range: [-108.2, 68.2] m, Z range: [-28.0, 34.9] m

Topic Name: /ars548/radar_front/detections
Timestamp: 1727234112.997
Total number of points: 462
X range: [3.6, 297.5] m, Y range: [-60.6, 71.2] m, Z range: [-27.6, 34.5] m

Topic Name: /ars548/radar_front/detections
Timestamp: 1727234113.048
Total number of points: 440
X range: [3.1, 297.0] m, Y range: [-64.2, 83.7] m, Z range: [-33.1, 31.6] m

Topic Name: /ars548/radar_front/detections
Timestamp: 1727234113.096
Total number of points: 456
X range: [3.5, 296.5] m, Y range: [-225.0, 104.6] m, Z range: [-24.3, 33.3] m

Topic Name: /ars548/radar_front/detections
Timestamp: 1727234113.146
Total number of points: 464
X range: [3.6, 295.7] m, Y range: [-161.2, 58.7] m, Z range: [-29.2, 26.7] m

[RUNNING] Bag Time: 1727234113.185758 Duration: 23.185523 / 59.999319
[RUNNING] Bag Time: 1727234113.186177 Duration: 23.185943 / 59.999319
[RUNNING] Bag Time: 1727234113.186246 Duration: 23.186012 / 59.999319
[RUNNING] Bag Time: 1727234113.187198 Duration: 23.186964 / 59.999319
[RUNNING] Bag Time: 1727234113.187260 Duration: 23.187025 / 59.999319
[RUNNING] Bag Time: 1727234113.187529 Duration: 23.187295 / 59.999319
[RUNNING] Bag Time: 1727234113.187972 Duration: 23.187737 / 59.999319
[RUNNING] Bag Time: 1727234113.188251 Duration: 23.188016 / 59.999319
[RUNNING] Bag Time: 1727234113.189036 Duration: 23.188801 / 59.999319
[RUNNING] Bag Time: 1727234113.189281 Duration: 23.189047 / 59.999319
[RUNNING] Bag Time: 1727234113.189377 Duration: 23.189142 / 59.999319
[RUNNING] Bag Time: 1727234113.189523 Duration: 23.189289 / 59.999319
[RUNNING] Bag Time: 1727234113.190671 Duration: 23.190437 / 59.999319
[RUNNING] Bag Time: 1727234113.190853 Duration: 23.190619 / 59.999319
[RUNNING] Bag Time: 1727234113.191063 Duration: 23.190829 / 59.999319
[RUNNING] Bag Time: 1727234113.191246 Duration: 23.191012 / 59.999319
[RUNNING] Bag Time: 1727234113.191338 Duration: 23.191103 / 59.999319
[RUNNING] Bag Time: 1727234113.191776 Duration: 23.191541 / 59.999319
[RUNNING] Bag Time: 1727234113.191988 Duration: 23.191753 / 59.999319
[RUNNING] Bag Time: 1727234113.192707 Duration: 23.192472 / 59.999319
[RUNNING] Bag Time: 1727234113.193826 Duration: 23.193592 / 59.999319
[RUNNING] Bag Time: 1727234113.194149 Duration: 23.193914 / 59.999319

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* /roscdistro: noetic
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ROS_MASTER_URI=http://laptop:11311/

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process[rosout-1]: started with pid [487]
started core service [/rosout]

```

Figure 5: Terminal output of pc_profiler.py for Radar (/ars548/radar_front/detections)

Task 3: Visualization of Rosbag Data

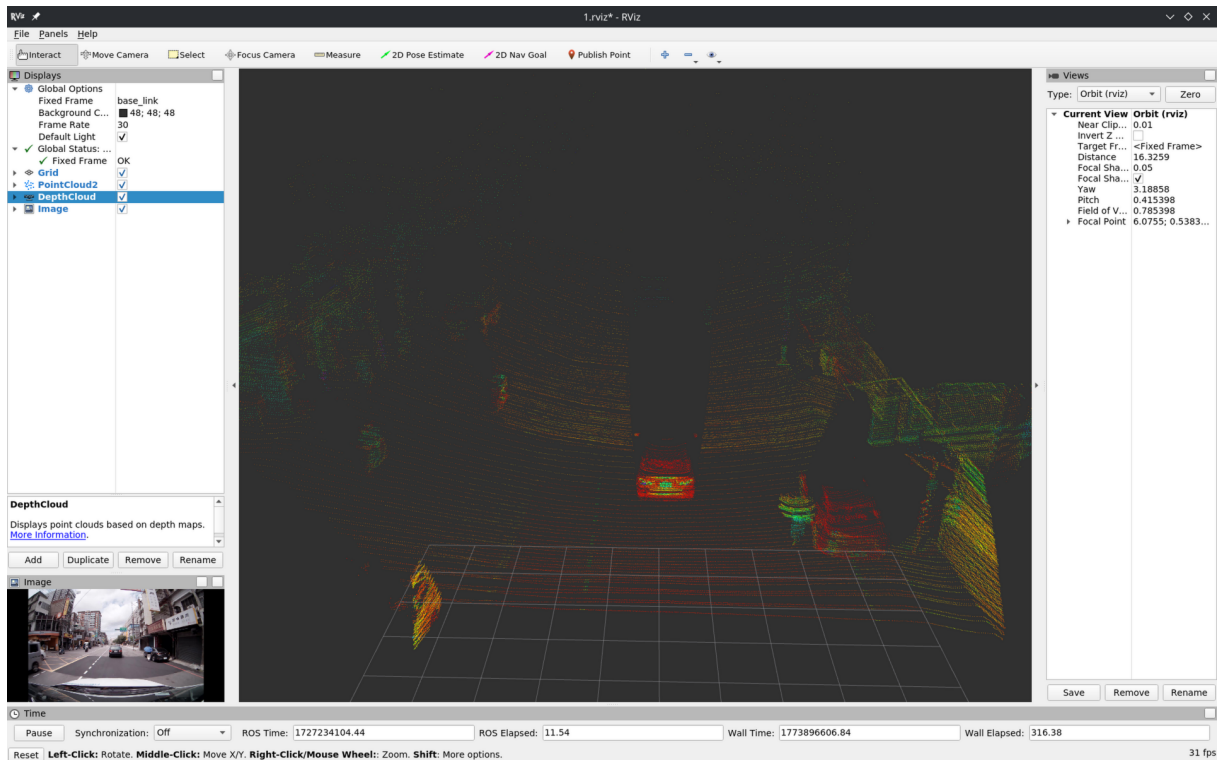


Figure 6: RViz pointcloud showcasing high-reflectivity objects alongside the camera image.

Task 4: Radar Relative Speed Visualization

YouTube Link: <https://youtu.be/jN5-tlbwZaI?si=e-uYSNdmzpDiPU11>

Task 5: How does LiDAR measure distance?

LiDAR (Light Detection and Ranging) measures distance using the **Time of Flight (ToF)** principle. The sensor emits a short laser pulse and simultaneously starts a timer. When the light reflects off an object and returns to a photodetector, the timer stops. The distance (d) is calculated using the constant speed of light ($c \approx 3 \times 10^8$ m/s) and the exact time interval (Δt) of the round trip:

$$d = \frac{c \cdot \Delta t}{2}$$

To achieve the centimeter-level accuracy needed for autonomous driving, this time interval must be measured with picosecond (10^{-12} s) precision. This is accomplished using a specialized integrated circuit called a **Time-to-Digital Converter (TDC)**. The laser firing triggers a high-speed "START" electrical signal, and the returning photons hitting the photodetector generate a "STOP" signal. The TDC acts as an ultra-fast hardware stopwatch, precisely measuring the microscopic gap between these two pulses and translating it into a digital distance value.